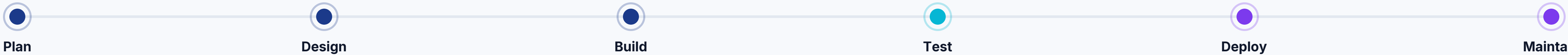


TECH STACK

- Python 3.12
- FastAPI
- Airflow 2.10
- SQLAlchemy
- NumPy
- Alembic
- WeasyPrint
- React 19
- TypeScript
- Tailwind
- Recharts
- PostgreSQL 16
- MySQL 8.4
- Argon2id + TOTP
- Server-Sent Events
- Docker Compose
- GitHub Actions
- pytest + mypy + Ruff

DEVELOPMENT TIMELINE



PROJECT OUTCOMES

- | | | | | |
|----------------------|------------------------------|------------------------|---------------------------|---------------------|
| Working pipeline | 28-table warehouse, 20 views | 12-rule DQ gate | REST API, 173 operations | 408 automated tests |
| 104 API smoke checks | React dashboard, 20 screens | Alembic-managed schema | 30 documents, 80 diagrams | PDF and CSV reports |